

4.3.17 Obviously $ARBA$ and $ASCP$ are rectangles.

$$\angle APQ = \angle AIP = 180^\circ - \angle AIB = 90^\circ - \frac{C}{2}$$

$$\tan \angle APS = \frac{AS}{AP} = \frac{AC \cos \frac{C}{2}}{AC \sin \frac{C}{2}} = \tan(90^\circ - \frac{C}{2})$$

$\therefore \angle APQ = \angle APS \rightarrow P, Q, S$ are collinear. Similarly P, Q, R are collinear.

